جامعة أبو بكر بلقايد
UNIVERSITY OF TLEMCEN

Sciences faculty

Computer Science Department

Computer Science Engineers 3rd year IA

Mobile applications

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Chapter 3:
Flutter installation & Setup

Objectives

- This chapter aims to:

  - ➢ List the prerequisites for installing Flutter and demonstrate the general steps to install it on a computer.

  - ➢ Configure an Android smartphone to test a Flutter app.

  - ➢ Identify the basic elements of a Flutter project and the role of its essential file and folder.

Plan

1. **Installation overview.**
2. **Installation on Windows.**
3. **Installation on macOS.**
4. **Use a real device (Android smartphone).**
5. **Flutter project.**

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Flutter installation overview

	On Windows	On macOS	On Linux
Build iOS Apps	**Not possible**	**Download & install Xcode** **Configure Xcode command-line tools** **Create local iOS simulator**	**Not possible**
Build Android Apps	**Download & install Android Studio** **Install SDK, command-line tools & build tools** **Create local Android emulator**		

- The code for all target platforms (iOS, Android, …) can be **written** on **any** machine (Linux, Windows, MacOs), but to build iOS & macOS apps the code can **only** be tested and run on macOS machines (the same for Widows and Linux apps).
- On the other hand, Android and web apps can be built on **all** OS.

Flutter installation overview

- **IDE for Flutter:**
  - Visual Studio Code (VS Code).
  - Android Studio.
  - Intellij IDEA.
  - FlutLab.io (online IDE with limited free plan and premium paid plan) : https://flutlab.io/pricing.

Flutter installation overview

- **Installation steps:**

Step 0 : IDE + extensions

a) **IDE :** for writing code.

b) **Extensions :** add Flutter and Dart plugins to the IDE to simplify the development process.

Step 1 : Flutter SDK

a) **Flutter SDK (UI Framework + collection of tools) :** for managing Flutter projects.

b) **Git :** version control software, used internally by Flutter SDK.

Step 2 : Platform Tools

a) **Android Studio :** used by Flutter SDK and needed for Android app deployment.

b) **Xcode:** used by Flutter SDK and needed for iOS app deployment.

Step 3 : Virtual Devices

a) **Android :** preview Flutter apps on Android Virtual Devices.

b) **iOS :** preview Flutter apps on virtual iOS devices.

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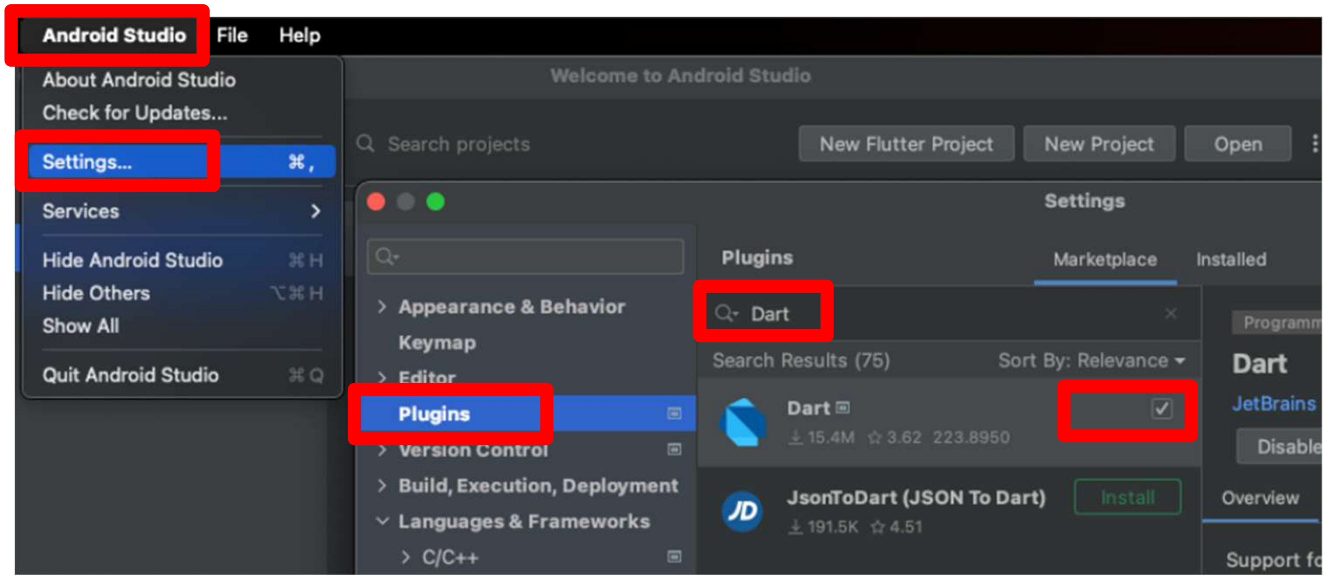
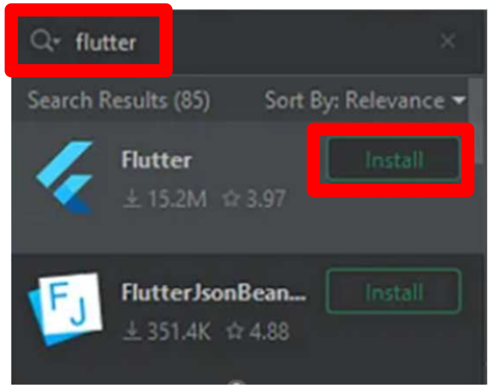
Flutter installation on Windows

- Step 0.a. install *VS Code & Android Studio* (see Steps.2.a.1/2.a.2)
- Step 0.b. add *Flutter & Dart* plugins (extensions).





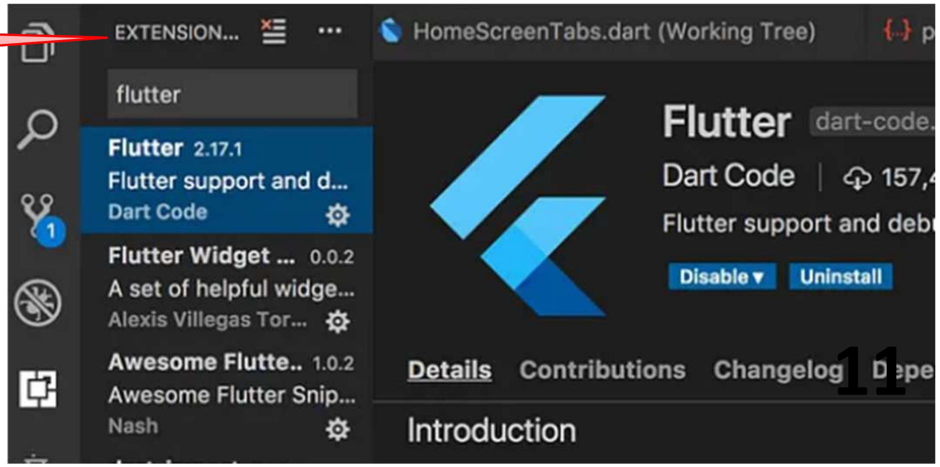
Add extensions to *Android Studio*

Flutter installation on Windows

- **Step 0.a.** install *VS Code & Android Studio* (see Steps.2.a.1/2.a.2)

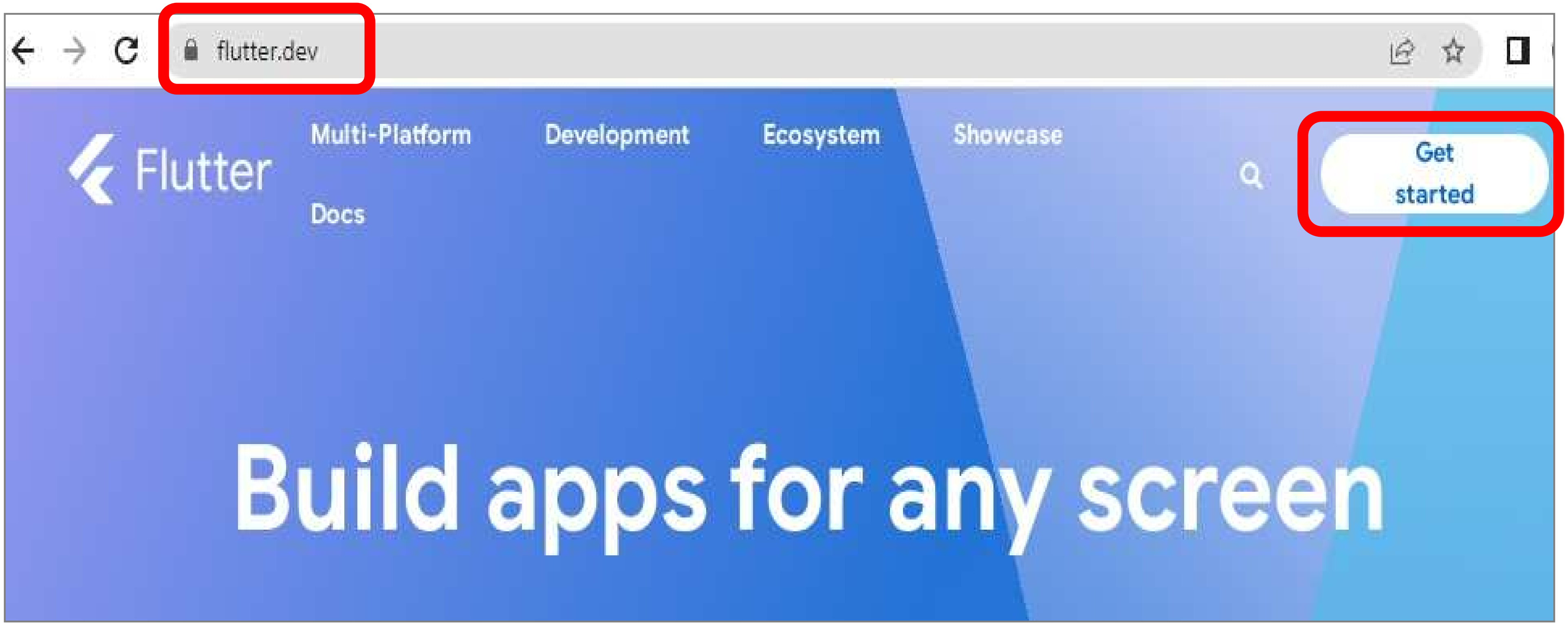
- **Step 0.b.** add *Flutter & Dart* plugins (extensions).

For VS Code, add the plugins from extension tab.



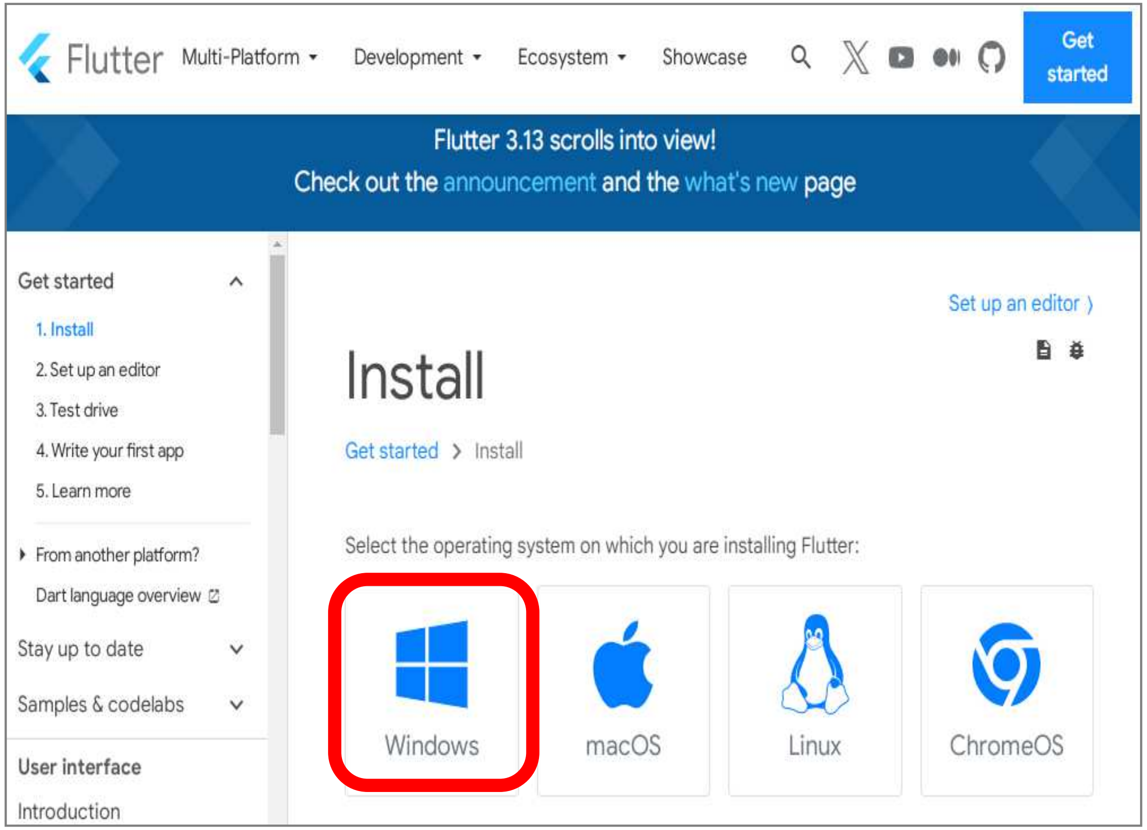
Flutter installation on Windows

- <u>Step 1.a.1.</u> Visit the official Flutter website ***Flutter.dev*** then click on ***Get started*** button :



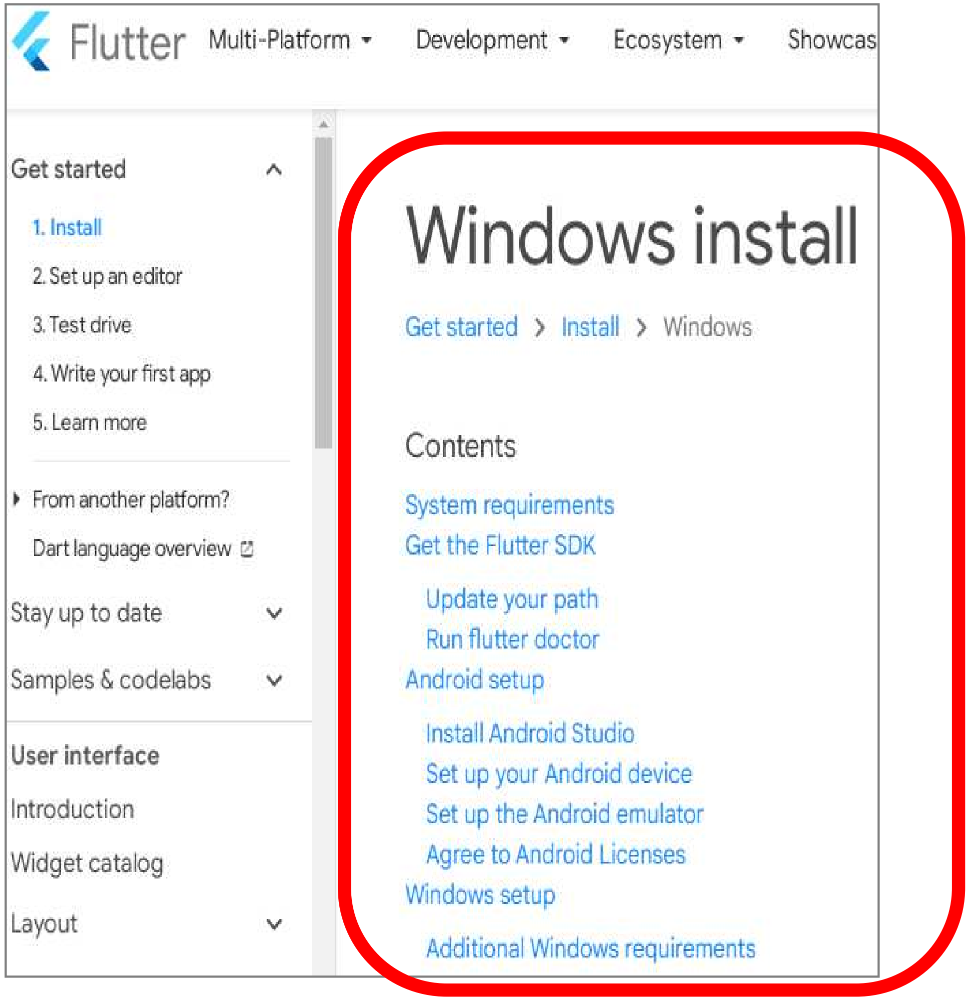
Flutter installation on Windows

- **Step 1.a.2.** Picking ***Windows*** as an option of the platform on which mobile apps will be developed :



Flutter installation on Windows

- <u>**Step 1.a.2.**</u> Picking ***Windows*** as an option of the platform on which mobile apps will be developed :



Flutter installation on Windows

- **Step 1.a.3.** System requirements: installing ***Git*** with the option "**use Git from the Windows Command Prompt"** :

System requirements

To install and run Flutter, your development environment must meet these minimum requirements:

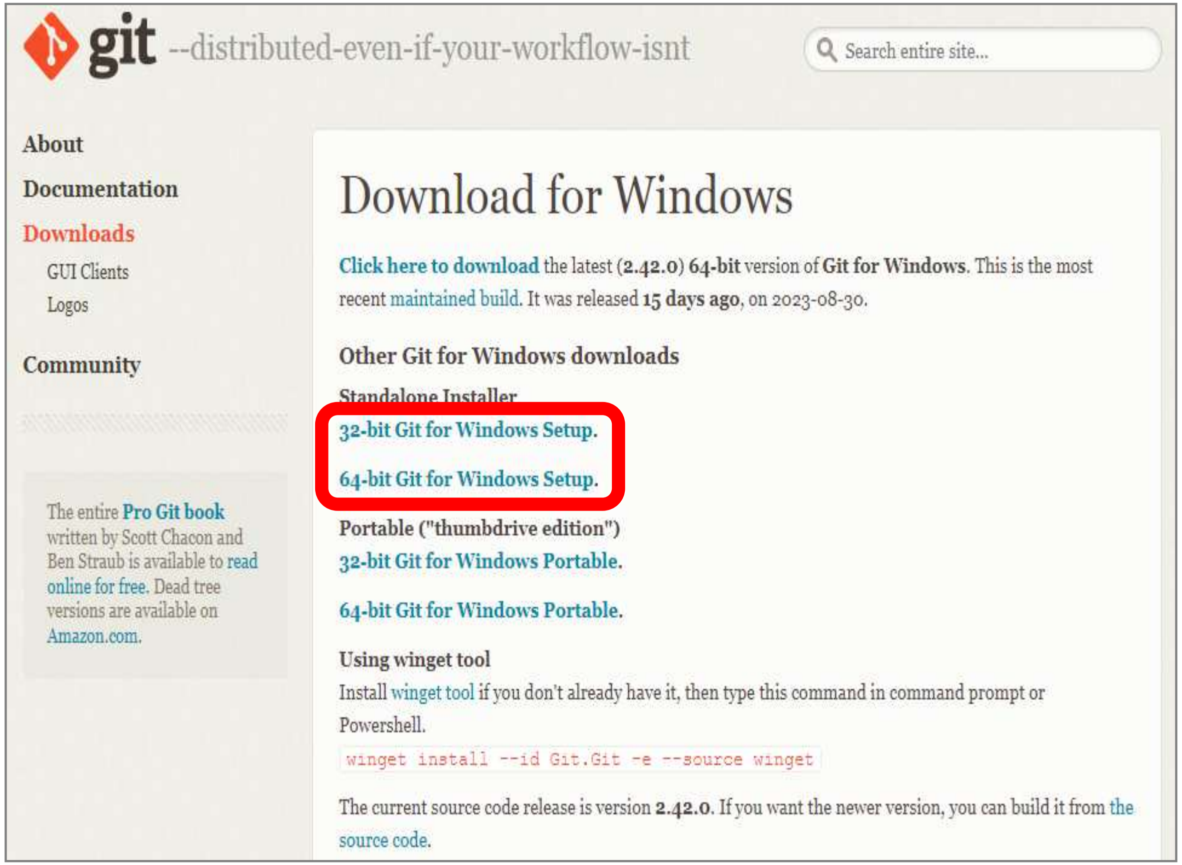
- **Operating Systems**: Windows 10 or later (64-bit), x86-64 based.
- **Disk Space**: 1.64 GB (does not include disk space for IDE/tools).
- **Tools**: Flutter depends on these tools being available in your environment.
  - Windows PowerShell 5.0 or newer (this is pre-installed with Windows 10)
  - Git for Windows 2.x, with the **Use Git from the Windows Command Prompt** option.

    If Git for Windows is already installed, make sure you can run `git` commands from the command prompt or PowerShell.

Follow the link to install Git for Windows

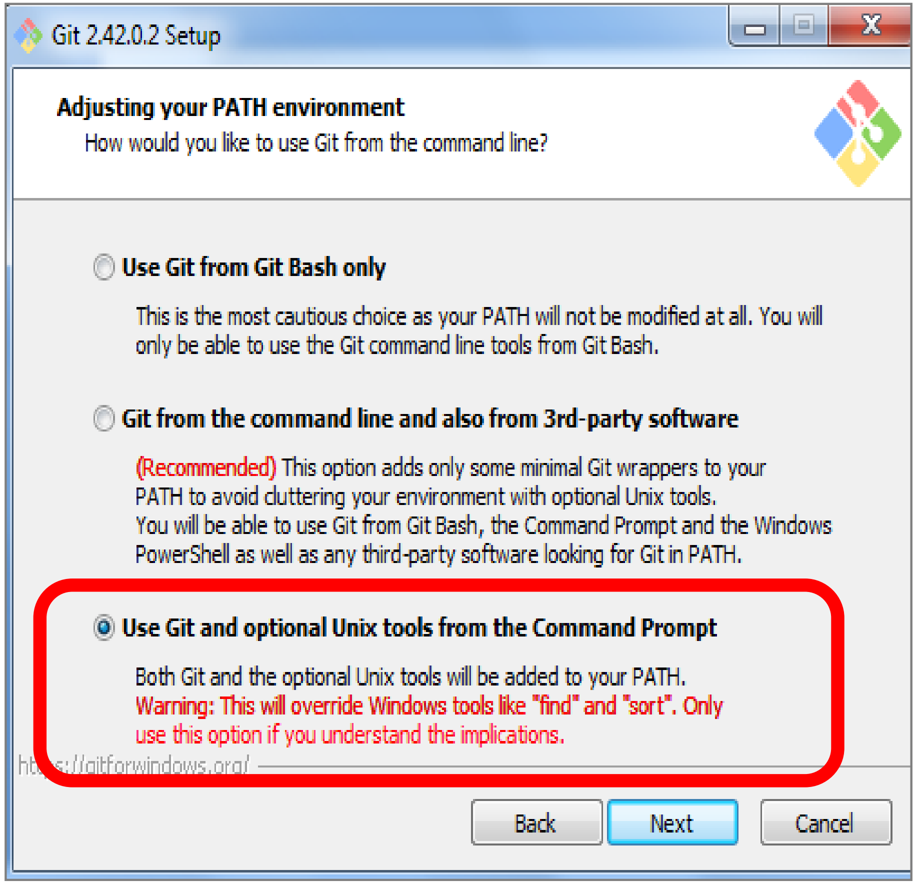
Flutter installation on Windows

- **Step 1.a.3.** System requirements: installing ***Git*** with the option "**use Git from the Windows Command Prompt**" :



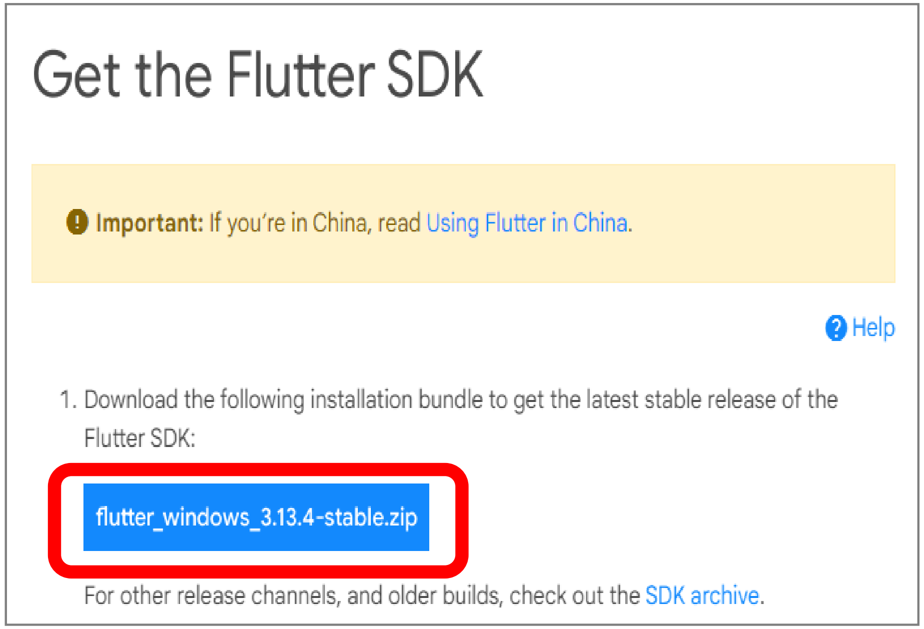
Flutter installation on Windows

- <u>Step 1.a.3.</u> System requirements: installing ***Git*** with the option "**use Git from the Windows Command Prompt"** :



Flutter installation on Windows

- **Step 1.b.1.** **Installing *Flutter*** itself (from the Flutter website):



i. **Download the zip file and unzip it.**
ii. **Move the flutter file somewhere else:**
e.g. to **C:\src\flutter** folder.

Flutter installation on Windows

- <u>Step 1.b.2.</u> **Update the path** : set environment variable.

Update your path

Help

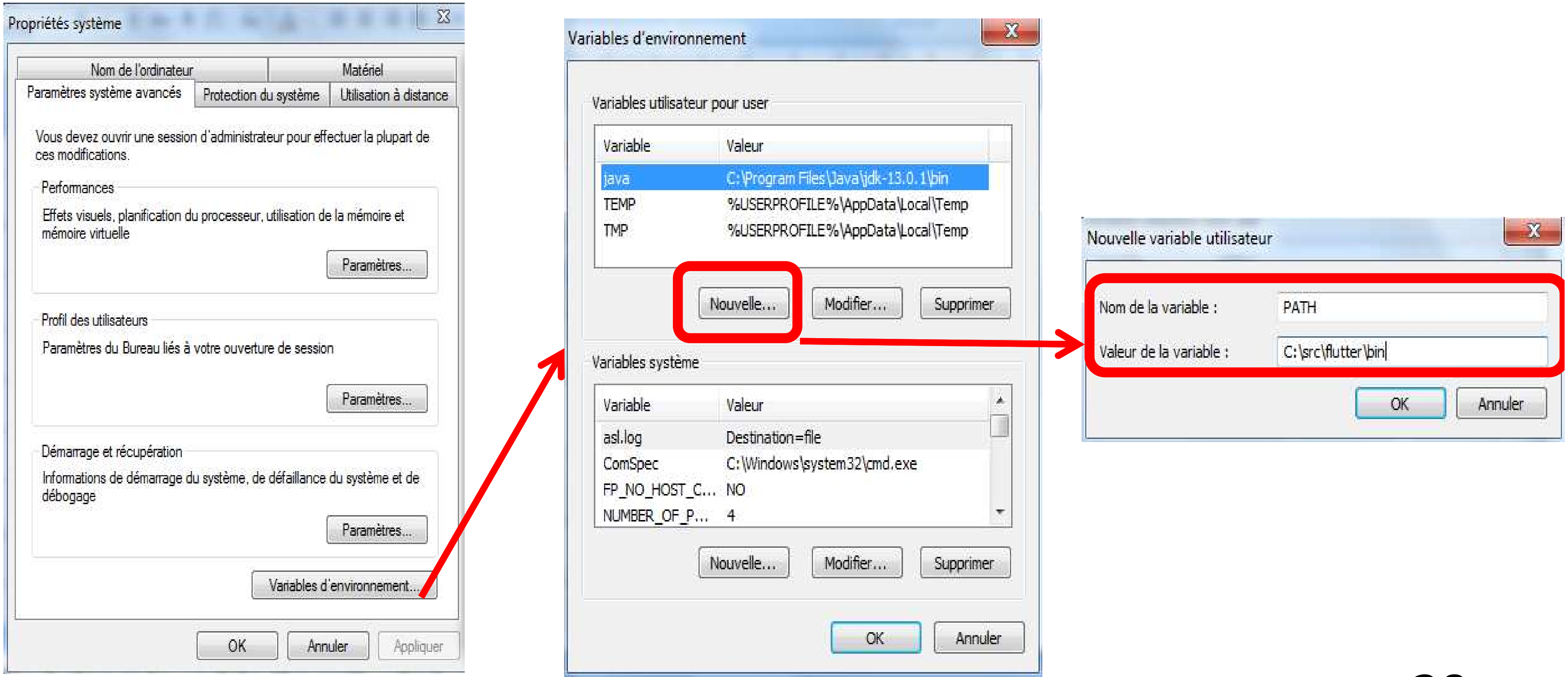
If you wish to run Flutter commands in the regular Windows console, take these steps to add Flutter to the `PATH` environment variable:

- From the Start search bar, enter 'env' and select **Edit environment variables for your account.**
- Under **User variables** check if there is an entry called **Path**:
  - If the entry exists, append the full path to `flutter\bin` using `;` as a separator from existing values.
  - If the entry doesn't exist, create a new user variable named `Path` with the full path to `flutter\bin` as its value.

You have to close and reopen any existing console windows for these changes to take effect.

Flutter installation on Windows

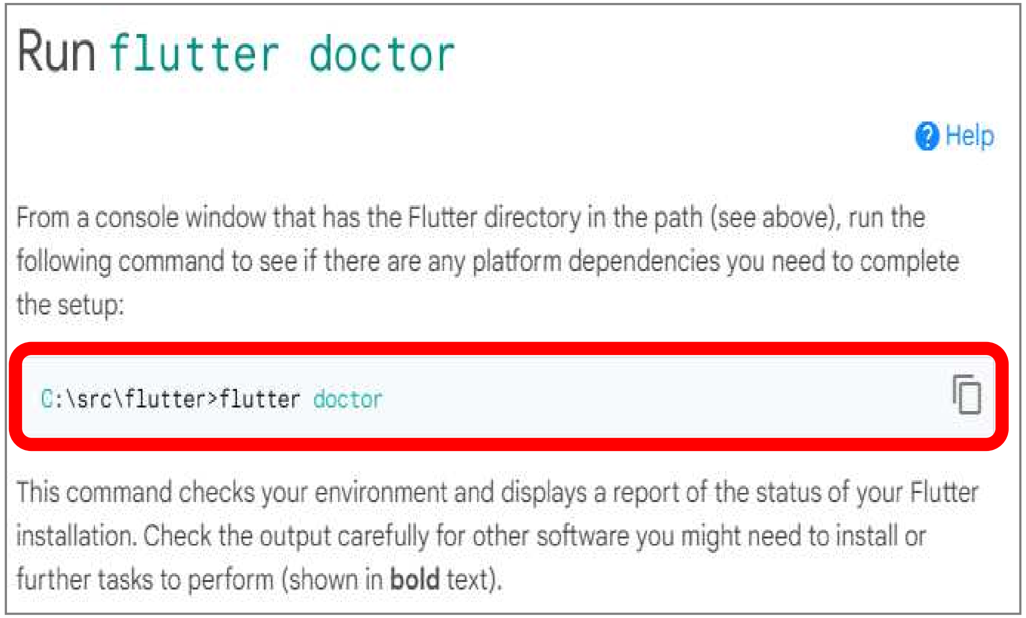
- **Step 1.b.2. Update the path** : set environment variable.



Look for "system environment variables" from the research bar of windows system.

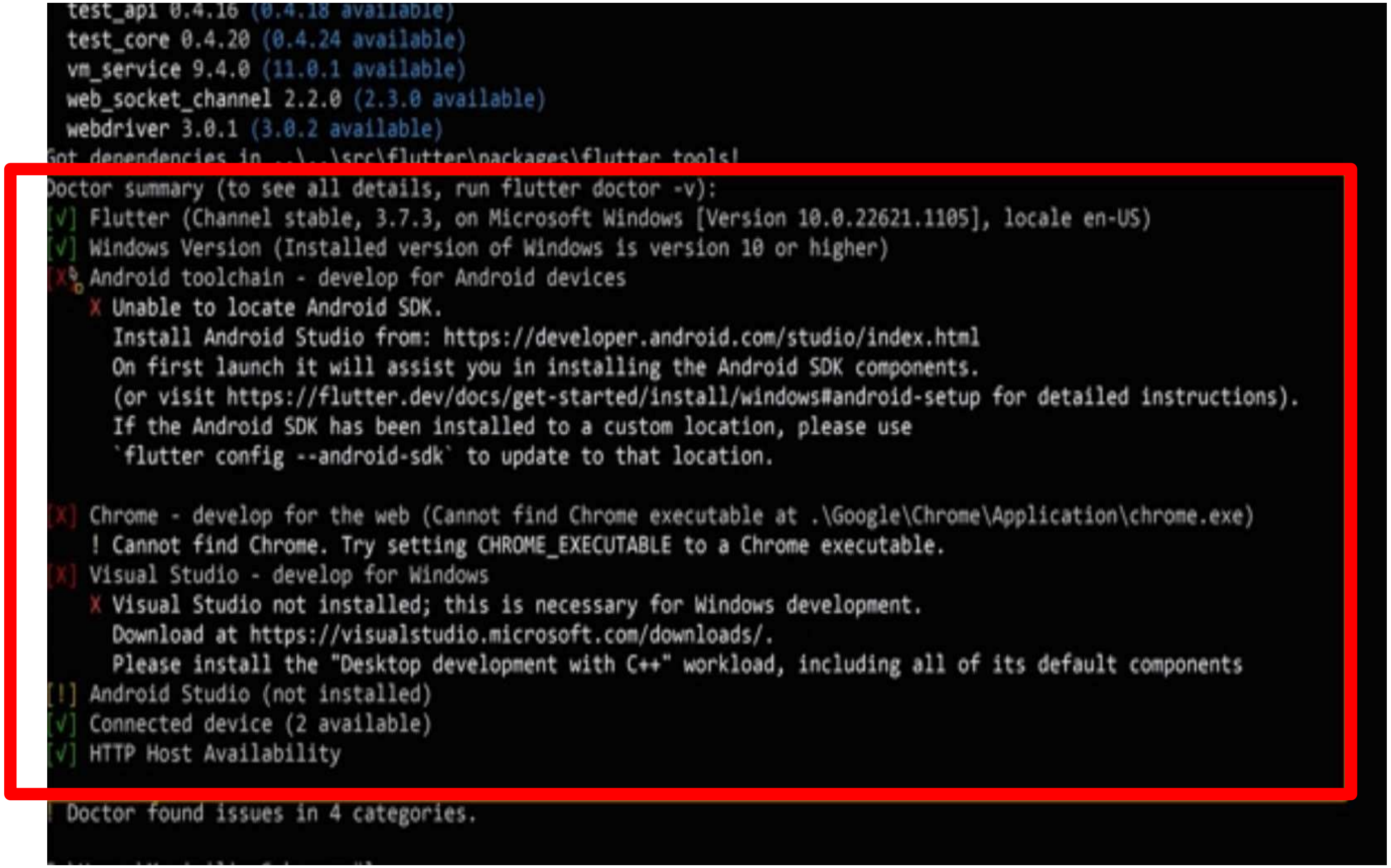
Flutter installation on Windows

- <u>Step 1.b.3.</u> Check platform dependencies to complete the setup : run ***flutter doctor*** command.



Flutter installation on Windows

- **Step 1.b.3.** Check platform dependencies to complete the setup : run ***flutter doctor*** command.



```
test_api 0.4.16 (0.4.18 available)
test_core 0.4.20 (0.4.24 available)
vm_service 9.4.0 (11.0.1 available)
web_socket_channel 2.2.0 (2.3.0 available)
webdriver 3.0.1 (3.0.2 available)
Got dependencies in ..\..\src\flutter\packages\flutter_tools!
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.7.3, on Microsoft Windows [Version 10.0.22621.1105], locale en-US)
[√] Windows Version (Installed version of Windows is version 10 or higher)
[X] Android toolchain - develop for Android devices
    X Unable to locate Android SDK.
      Install Android Studio from: https://developer.android.com/studio/index.html
      On first launch it will assist you in installing the Android SDK components.
      (or visit https://flutter.dev/docs/get-started/install/windows#android-setup for detailed instructions).
      If the Android SDK has been installed to a custom location, please use
      `flutter config --android-sdk` to update to that location.

[X] Chrome - develop for the web (Cannot find Chrome executable at .\Google\Chrome\Application\chrome.exe)
    ! Cannot find Chrome. Try setting CHROME_EXECUTABLE to a Chrome executable.
[X] Visual Studio - develop for Windows
    X Visual Studio not installed; this is necessary for Windows development.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[!] Android Studio (not installed)
[√] Connected device (2 available)
[√] HTTP Host Availability

! Doctor found issues in 4 categories.
```

Flutter installation on Windows

- Step 2.a. Prepare the system for Android app development :

Android setup

Note: Flutter relies on a full installation of Android Studio to supply its Android platform dependencies. However, you can write your Flutter apps in a number of editors; a later step discusses that.

Step 0.a.

Install Android Studio

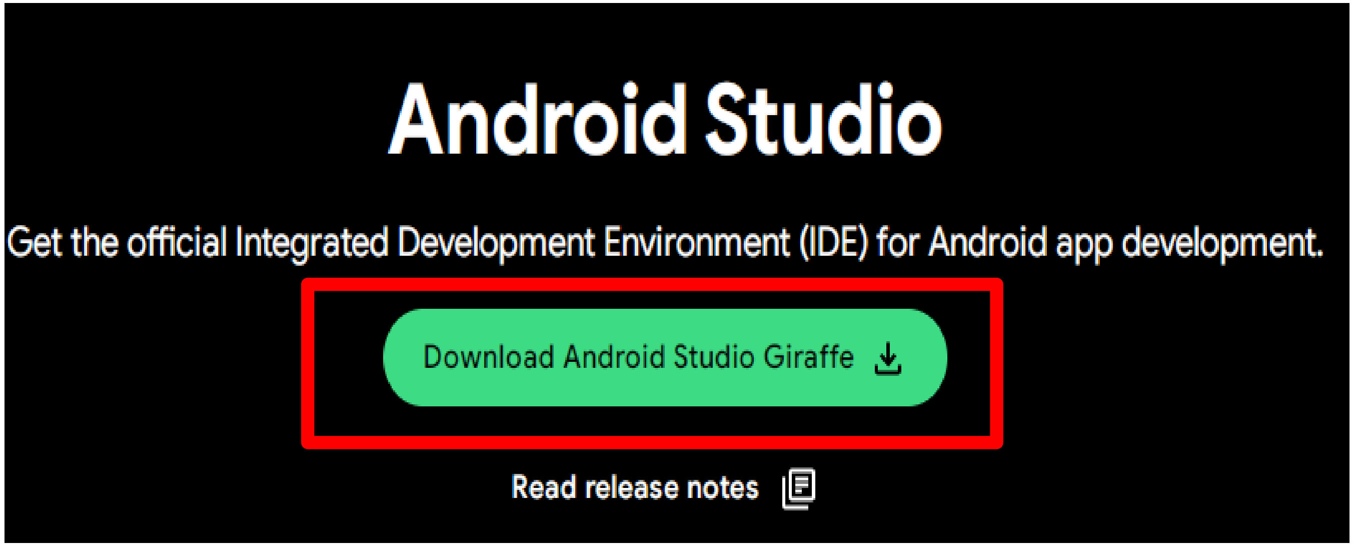
Help

1. Download and install Android Studio.
2. Start Android Studio, and go through the 'Android Studio Setup Wizard'. This installs the latest Android SDK, Android SDK Command-line Tools, and Android SDK Build-Tools, which are required by Flutter when developing for Android.
3. Run `flutter doctor` to confirm that Flutter has located your installation of Android Studio. If Flutter cannot locate it, run `flutter config --android-studio-dir=<directory>` to set the directory that Android Studio is installed to.

Flutter installation on Windows

- Step 2.a. Prepare the system for Android app development :

Step 0.a. → Step 2.a.1. Download Android Studio (*) :

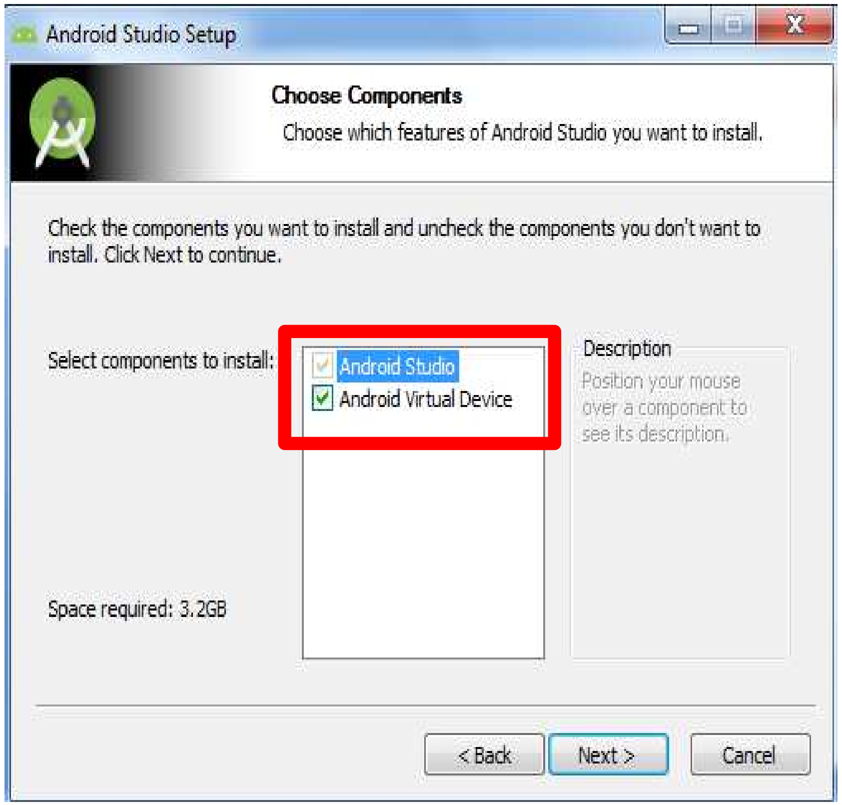


(*) : Google's official Android development environment.

Flutter installation on Windows

- **Step 2.a.** Prepare the system for Android app development :

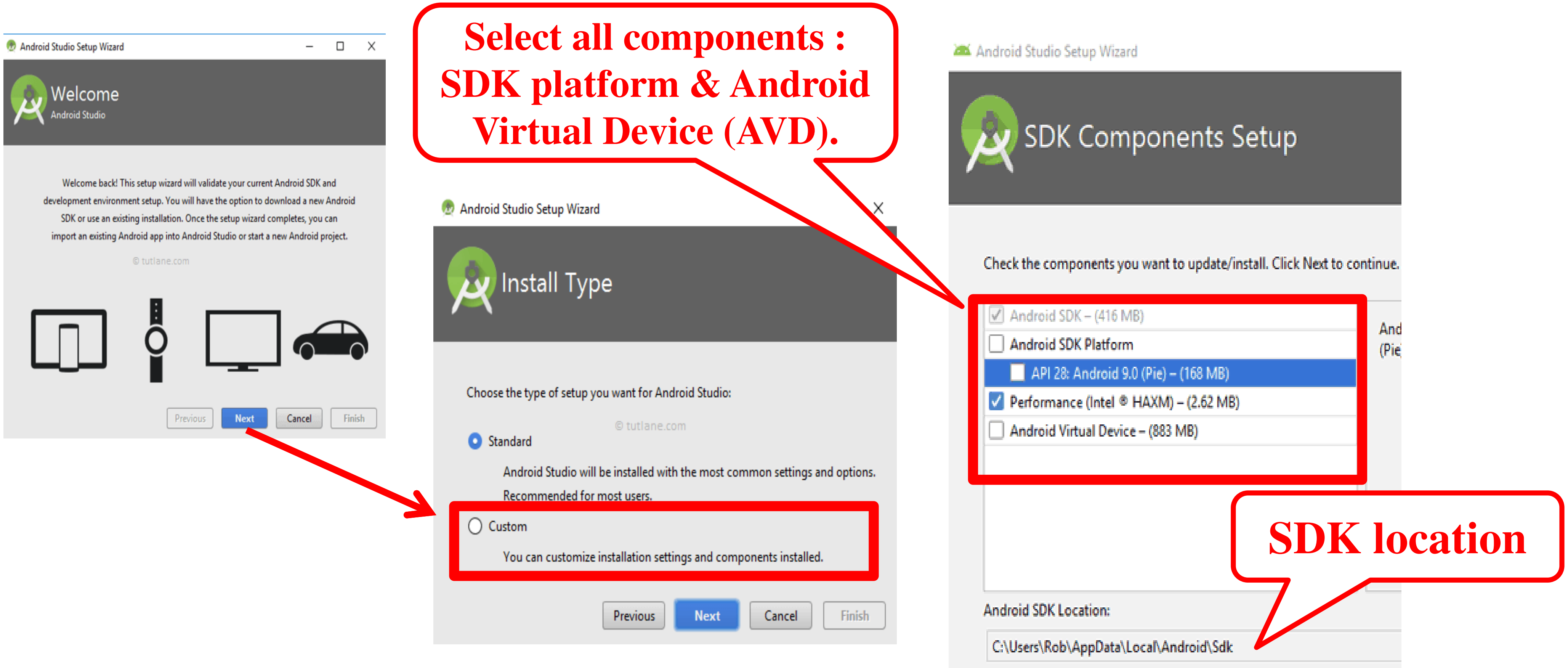
Step 0.a. → **Step 2.a.2.** Install Android Studio :



Flutter installation on Windows

- **Step 2.a.** Prepare the system for Android app development :

  Step 2.a.3. Start and setup Android Studio: **SDK Manager**



Flutter installation on Windows

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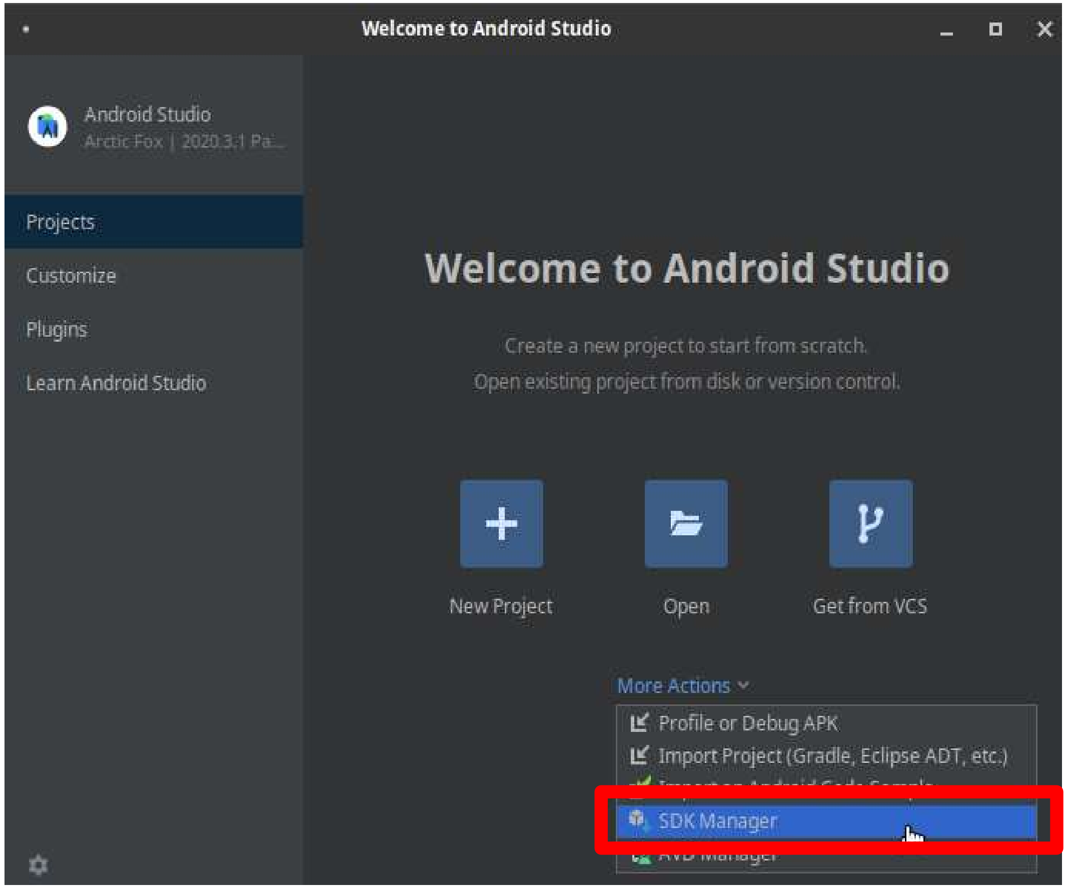

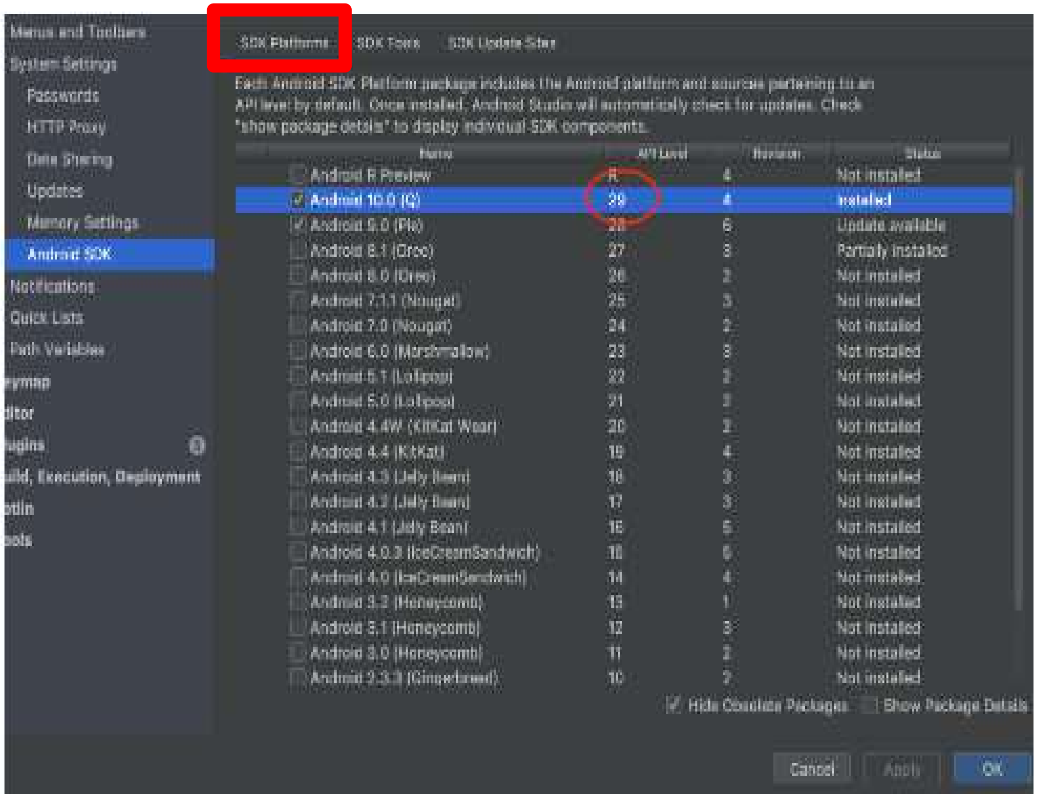

Flutter installation on Windows

- **Step 2.a.** Prepare the system for Android app development :

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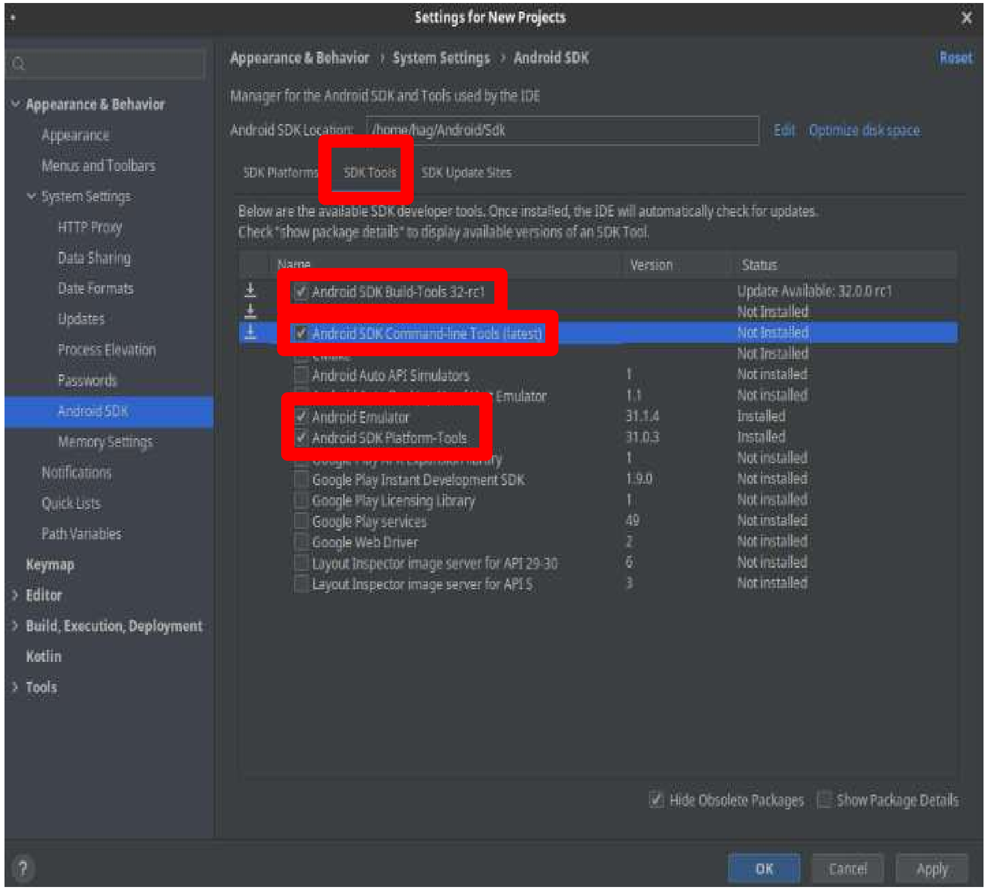
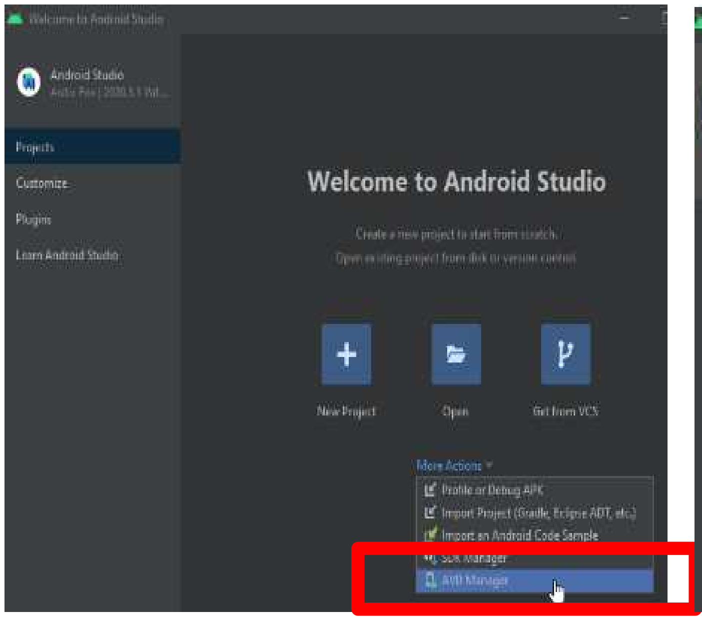
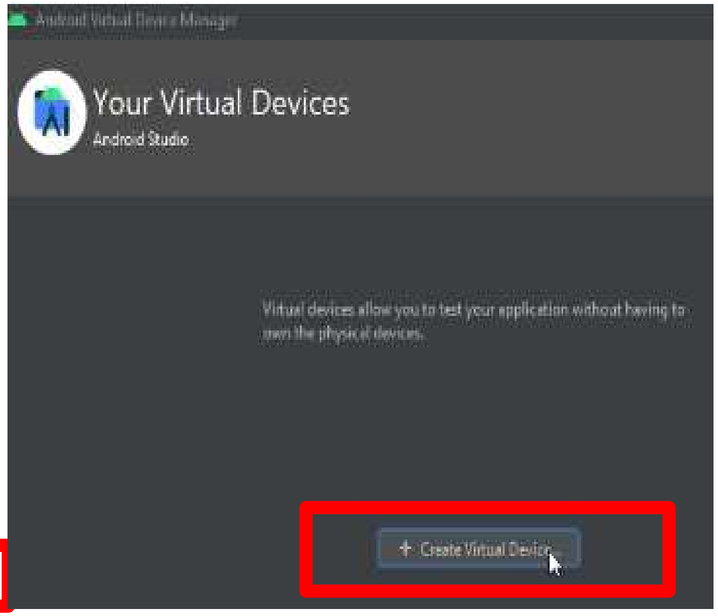
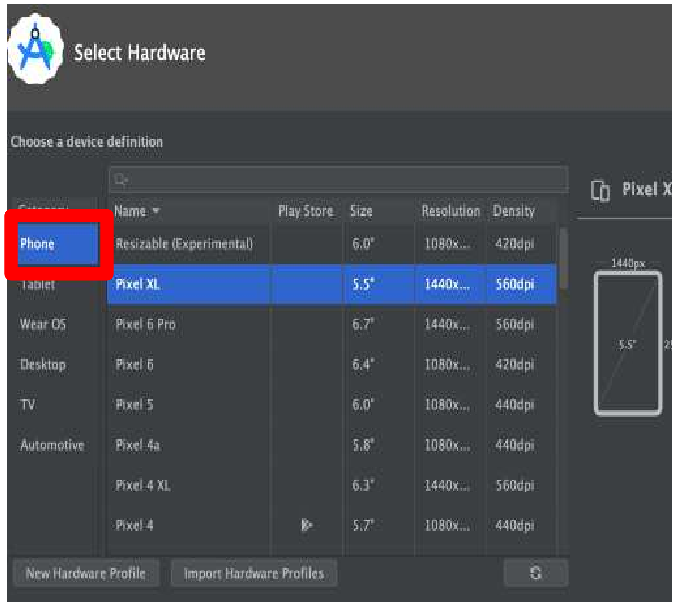


Flutter installation on Windows

- <u>**Step 3.a.**</u> Setup the ***Virtual Device Manager*** to create an emulator.

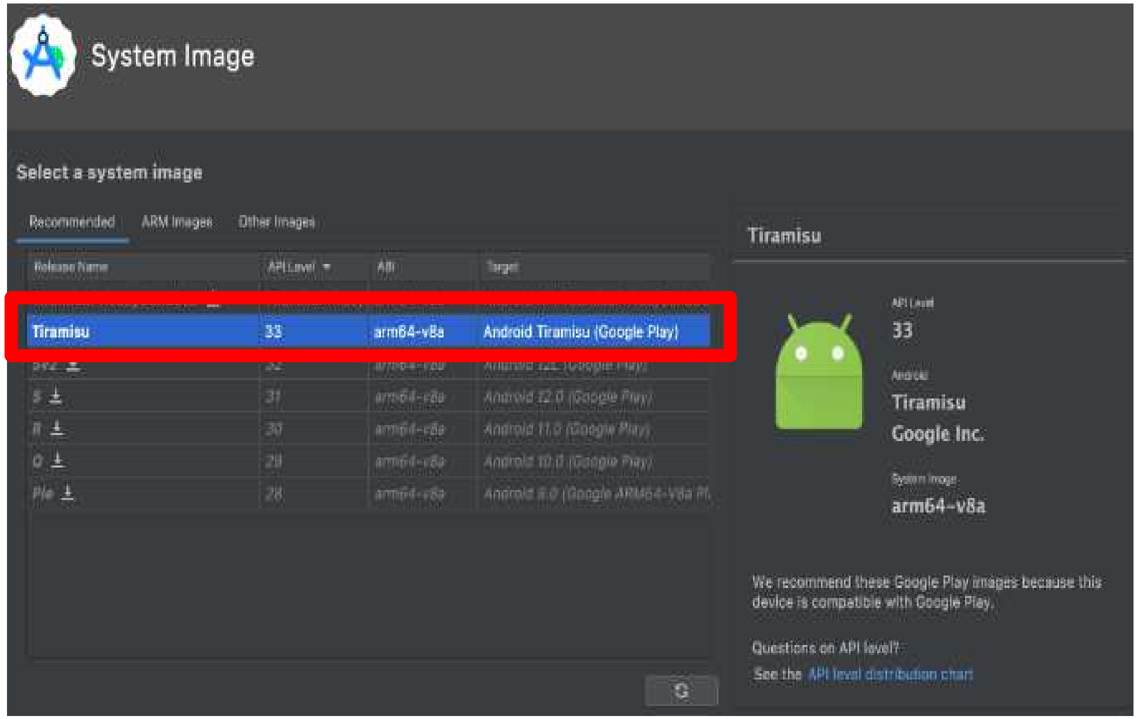


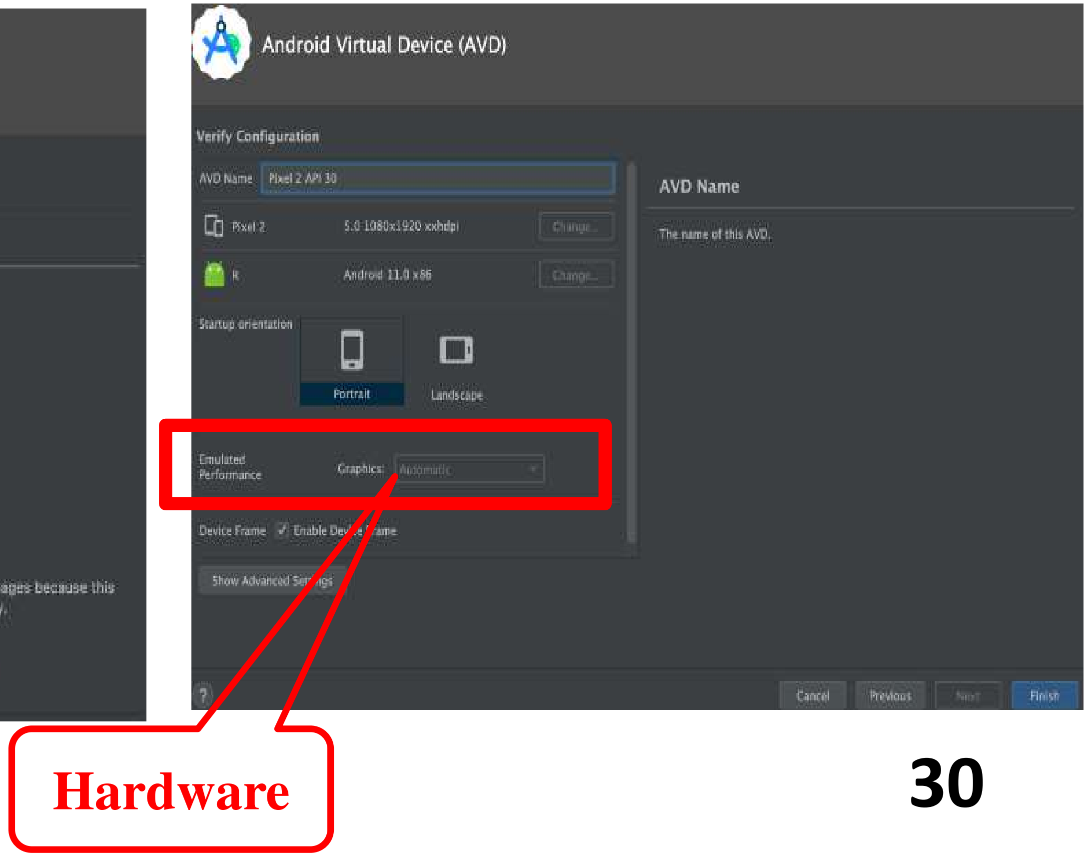




Flutter installation on Windows

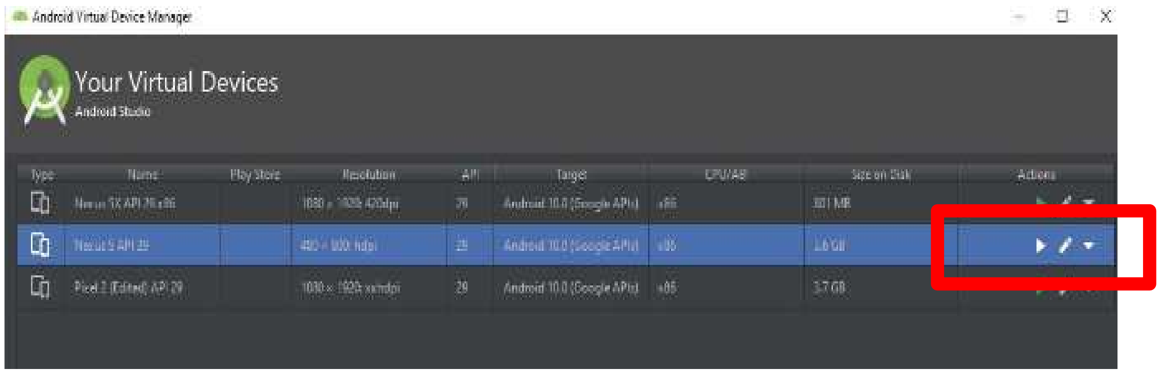
- <u>**Step 3.a.**</u> Setup the ***Virtual Device Manager*** to create an emulator.





Flutter installation on Windows

- **Step 3.a.** Setup the ***Virtual Device Manager*** to create an emulator.



- **Remark:** is it also possible to launch the emulator from VS Code.

Flutter installation on Windows

- For more details on
    - IDE installation of Android Studio and Visual Studio Code (including Flutter & Dart extensions) and,
    - Running Flutter project on Android emulator (AVD).

- Please get a look at the videos :
    - DemoFlutterInstallation.mp4 (*)
    - DemoFlutterInstallation2.mp4 (*)

(*) Read videos with VLC media player.

Plan

1. Installation overview.
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3. **Installation on macOS.**
4. Use a real device (Android smartphone).
5. Flutter project.

Flutter installation on macOS

- **Installing Flutter on macOS:** for Flutter installation process on macOS system and running Flutter project on iOS simulator, please get a look at the video :

    - DemoFlutterInstallation4macOS.mp4 (*)

(*) Read videos with VLC media player.

Plan

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5. Flutter project.

Use a real device

- Running a Flutter project on an ***Android smartphone:***
  1) Activate developer options by clicking 7 times on the build number from « settings/About phone … »
  2) Activate USB debugging option on the phone from « settings/developer options ».
  3) Connect physical device using USB cable.
  4) Launch the app from the IDE (Android Studio, VS code) .

Plan

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5. **Flutter project.**

Flutter project

- **Creation:** four options exist for creating a new Flutter project.

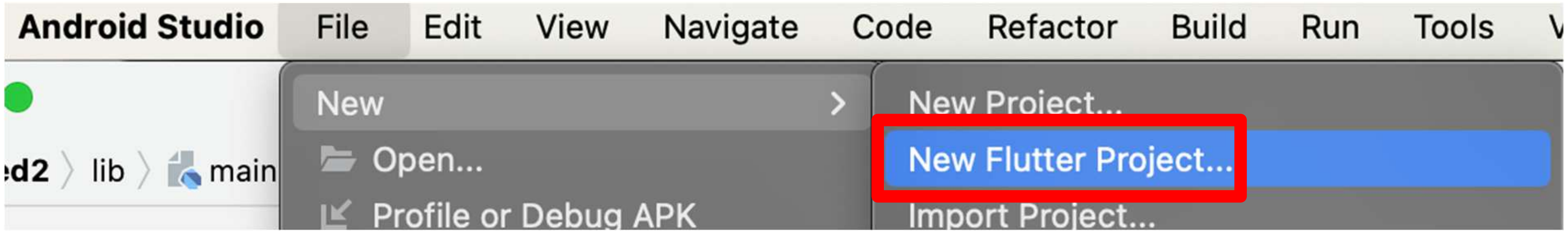
  1. From the windows prompt command:

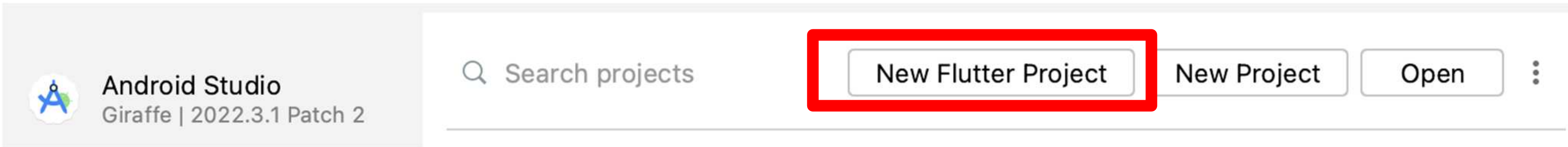
! Flutter Project name must not contain capital letters.

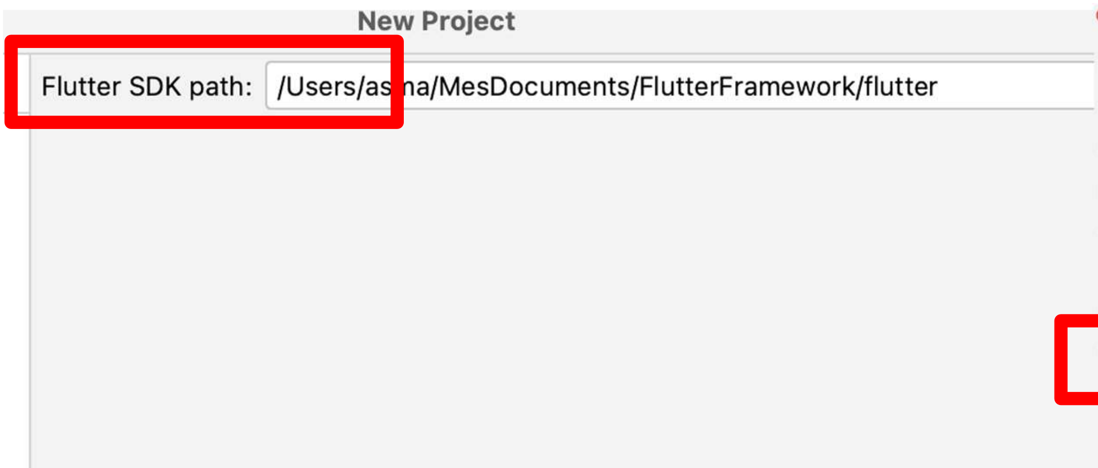
Flutter project

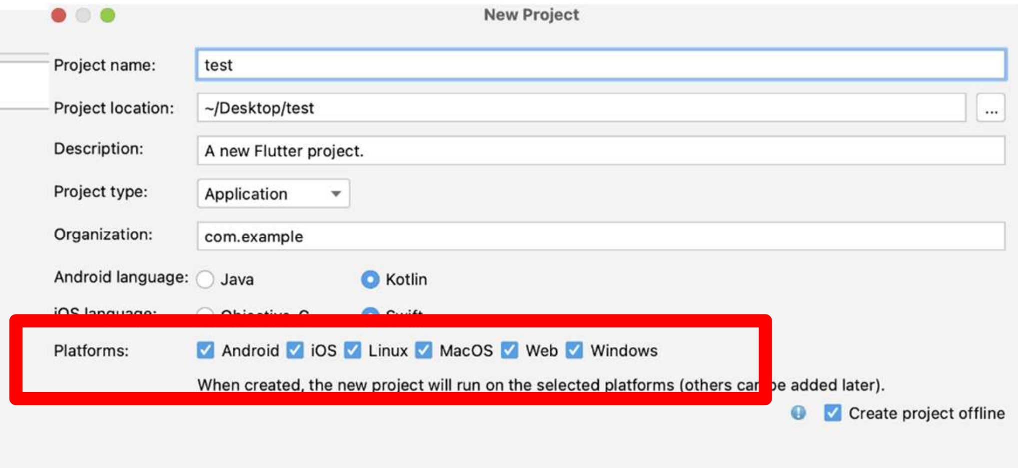
- **Creation:** four options exist for creating a new Flutter project.

  2. From Android Studio:





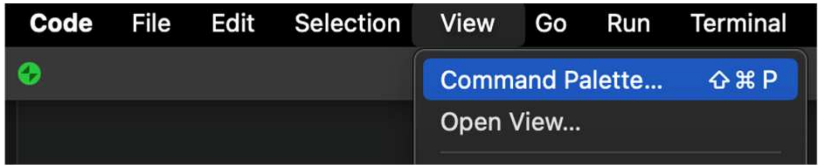


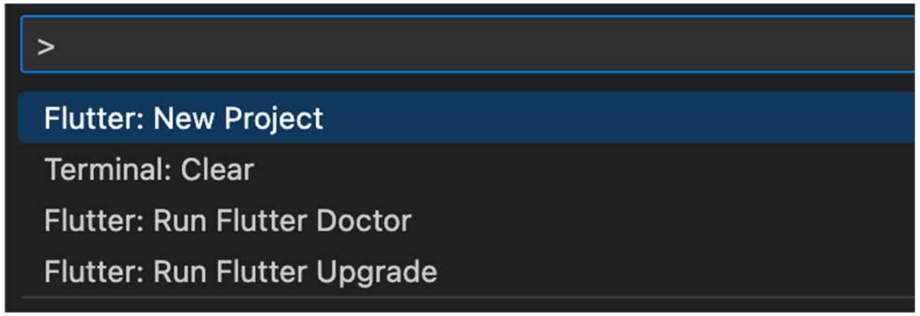


Flutter project

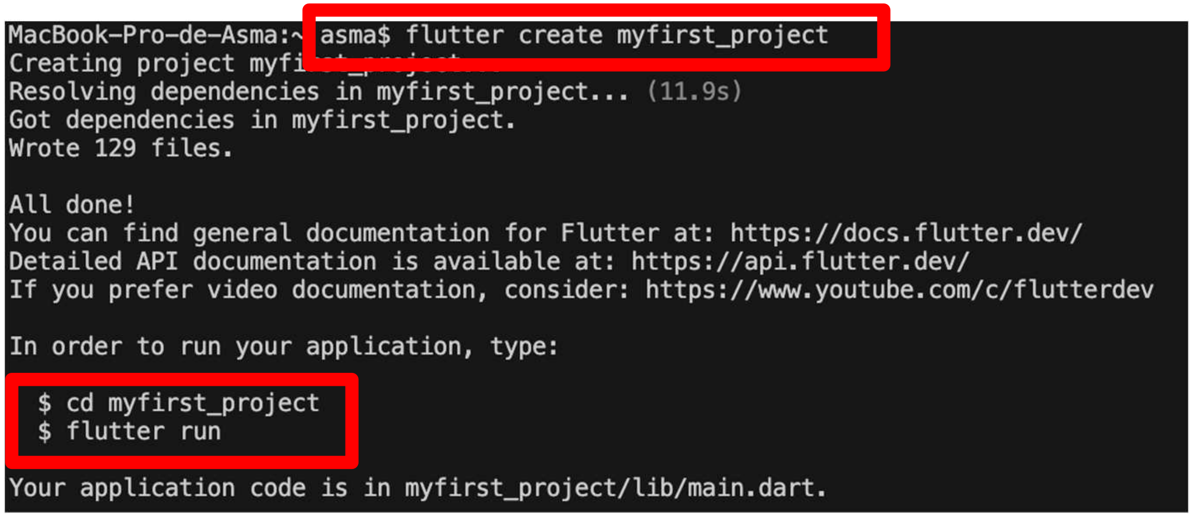
- **Creation:** four options exist for creating a new Flutter project.

    3. From the VS Code command palette:



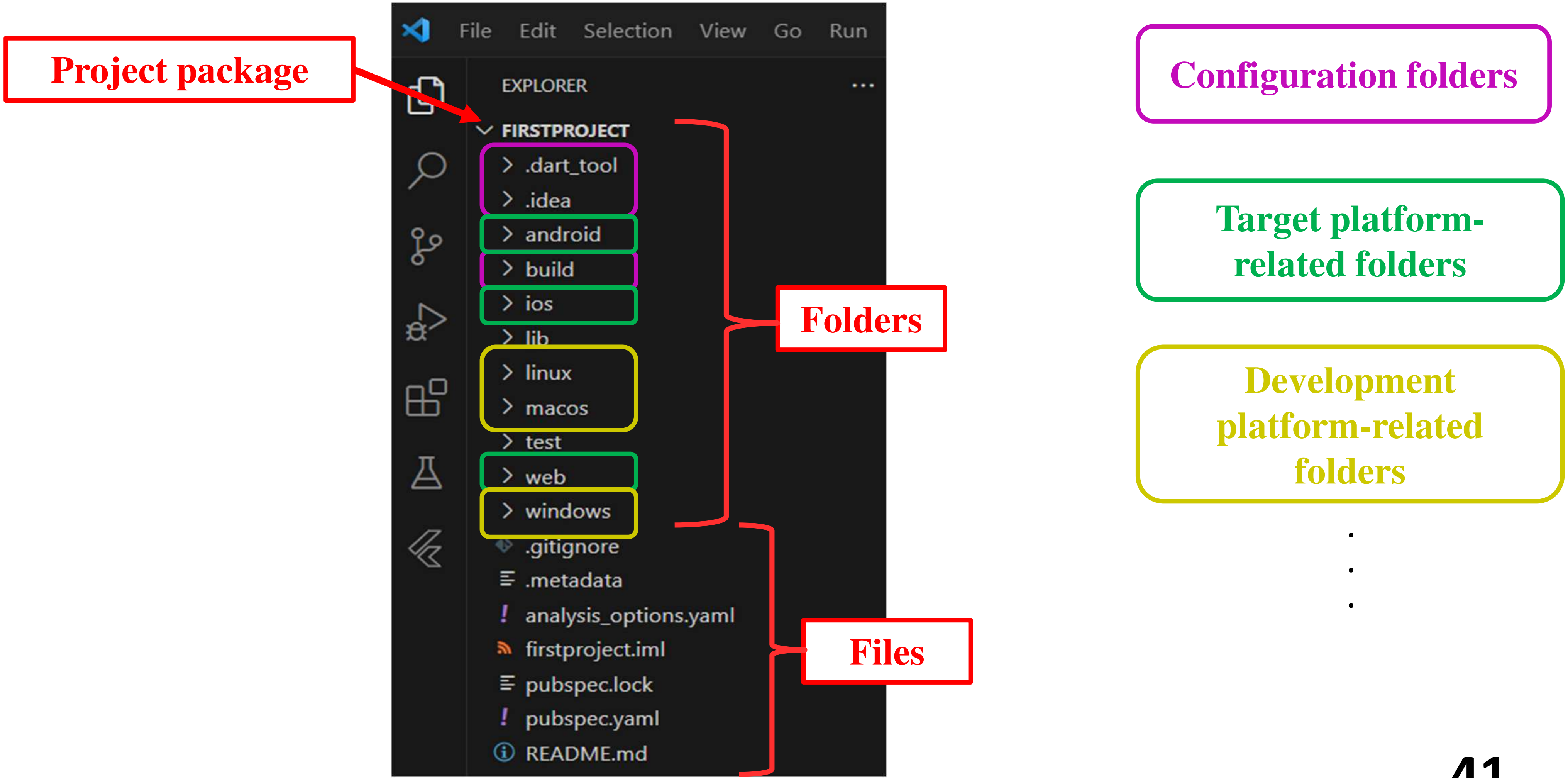


    4. From the VS Code terminal:



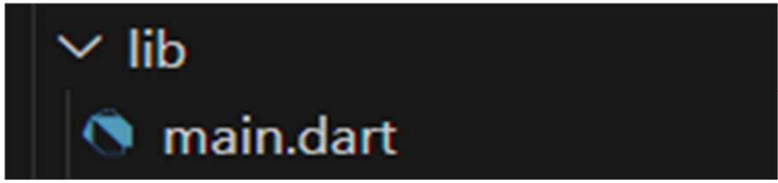
Flutter project

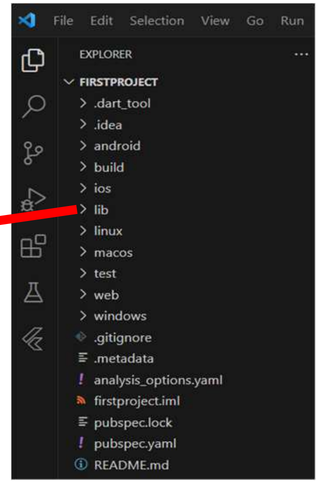
- **Structure:**



Flutter project

- **Structure: "lib folder"**
  - Contains Flutter app code (files end with .dart).
  - Contains **main.dart** (the entry point of the Flutter app).





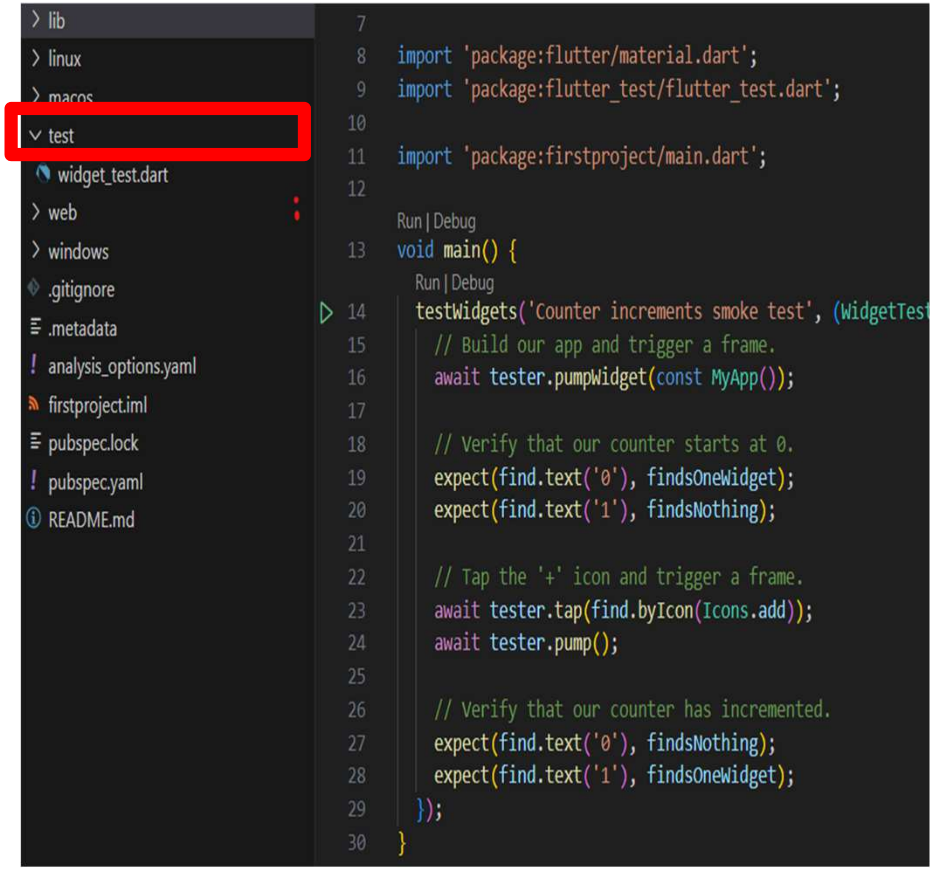
  - **Example:** basic Flutter app (Hello World!)

```
import 'package:flutter/widgets.dart'; // basic set of widgets
// When Dart is running the application, it calls to the main() function
 main() => runApp( // The function runApp() starts the Flutter application
Text ( // this is a widget, it renders the given text
   'Hello, World!!!', // the first argument is a text that needs to be rendered
   textDirection: TextDirection.ltr, // here we set the direction "left to right"
), );
```

Flutter project

- **Structure: "test folder"**

  - Contains Dart code files used to test the application code.
  - Test files are used when "flutter –test" command is run.
  - Test files should have the suffix " _test.dart".



Flutter project

- **Structure: "pubspec.yaml file"**
  - Used to add third party packages to the project (e.g. include images into project).
  - Specifies project descriptions, constraints, dependencies, version, assets, …

.gitignore
.metadata
analysis_options.yaml
firstproject.iml
pubspec.lock
pubspec.yaml
README.md

Add custom fonts

firstproject > pubspec.yaml

```yaml
name: firstproject
description: A new Flutter project.
publish_to: 'none'
version: 1.0.0+1
environment:
  sdk: '>=3.1.3 <4.0.0'
dependencies:
  flutter:
    sdk: flutter
    cupertino_icons: ^1.0.2

dev_dependencies:
  flutter_test:
    sdk: flutter
  flutter_lints: ^2.0.0

flutter:
  uses-material-design: true
  assets:
   - images/a_dot_burr.jpeg
   fonts:
     - family: Schyler
         fonts:
       - asset: fonts/Schyler-Regular.ttf
       - asset: fonts/Schyler-Italic.ttf
         style: italic
     - family: Trajan Pro
       fonts:
         - asset: fonts/TrajanPro.ttf
         - asset: fonts/TrajanPro_Bold.ttf
           weight: 700
```

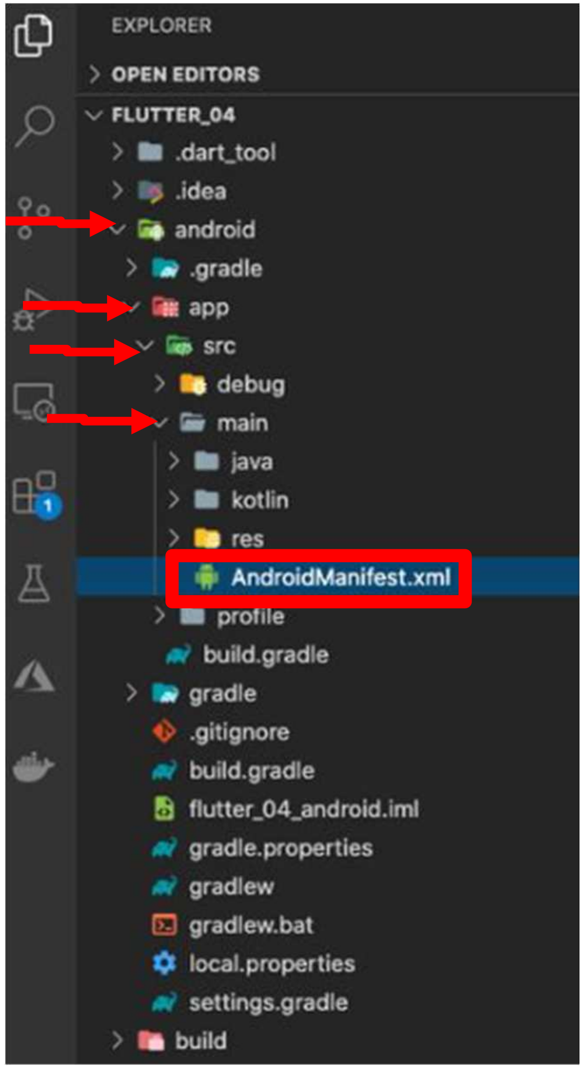
Flutter project

- **Structure: "AndroidManifest.xml file"**
  - Describes the essential information about the app to the Android build tools, Android OS, and Google Play.
  - It is used to configure the Android-specific settings of the app such as:
    - **Permissions** the app requires (camera, internet,…).

```
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.CAMERA" />
```

    - Declare **hardware** or **software** features the app requires (Bluetooth, GPS, camera,…).

```
<uses-feature android:name="android.hardware.camera" android:required="true" />
```



Flutter project

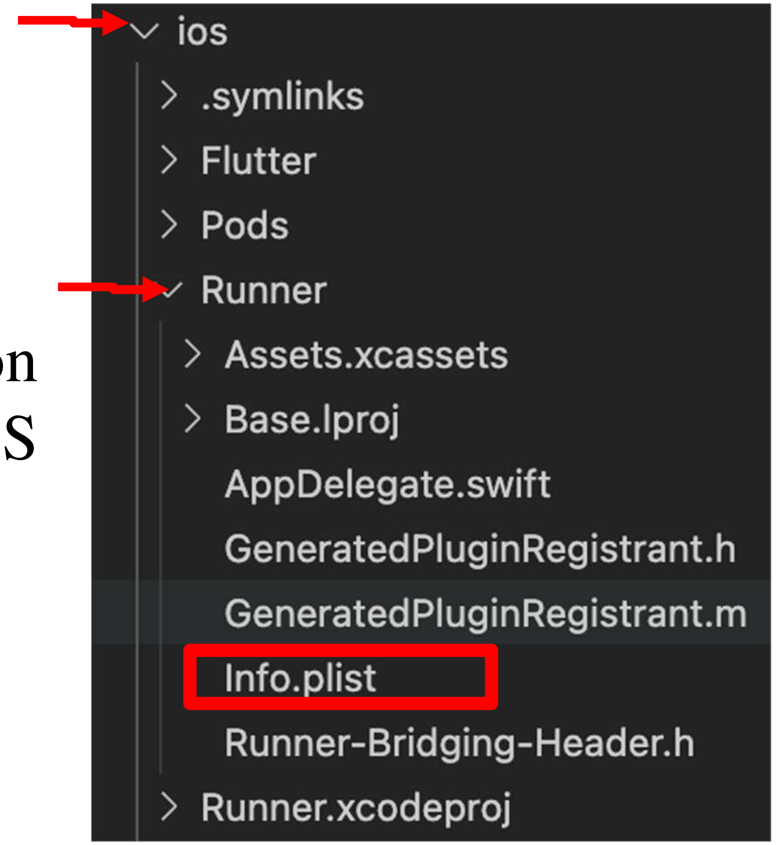
- **Structure: "Info.plist file"**

➢ The "*Info.plist"* file is the iOS equivalent of the *Android AndroidManifest.xml.*

➢ It contains essential configuration information about the app to the iOS build and the system.

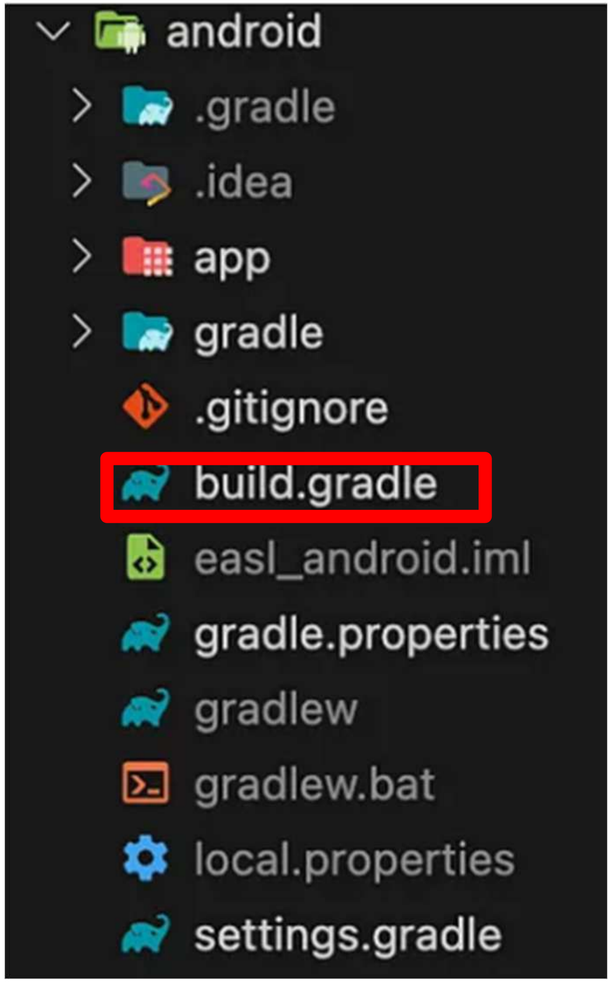
➢ Defines:

- App name and bundle identifier,
- Version and permissions,
- Supported device orientations,
- Other system-related settings.



Flutter project

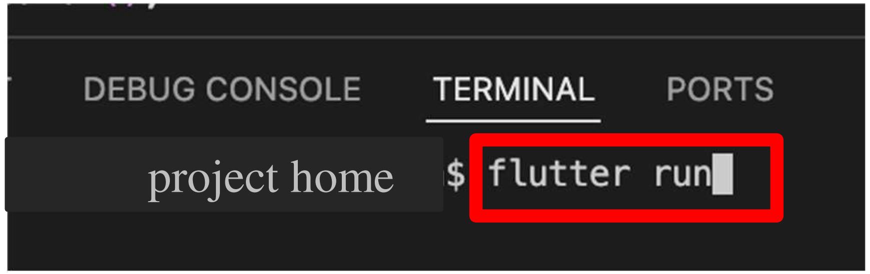
- **Structure: "build.gradle"**
  - ➢ Used for managing and building the Android-specific aspects of the Flutter app.
  - ➢ It is responsible for building automatically the Android APK.
  - ➢ Il allows managing any native Android dependencies or plugins that the Flutter app may require on Android (e.g. firebase_messaging or google_maps_flutter).



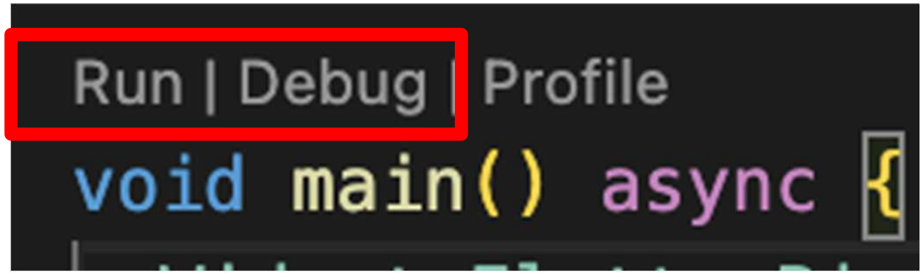
Flutter project

- **Run the app:** four options exist for running a Flutter project.

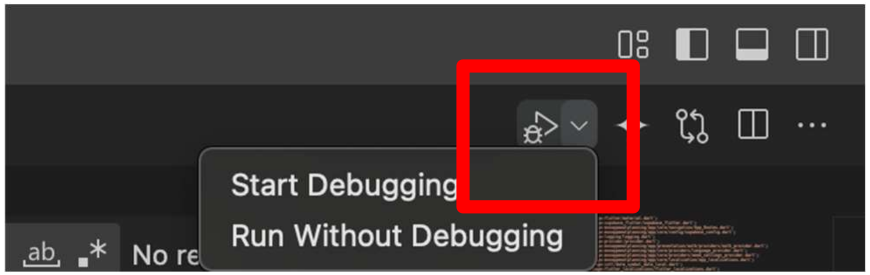
1. From the VS Code terminal:



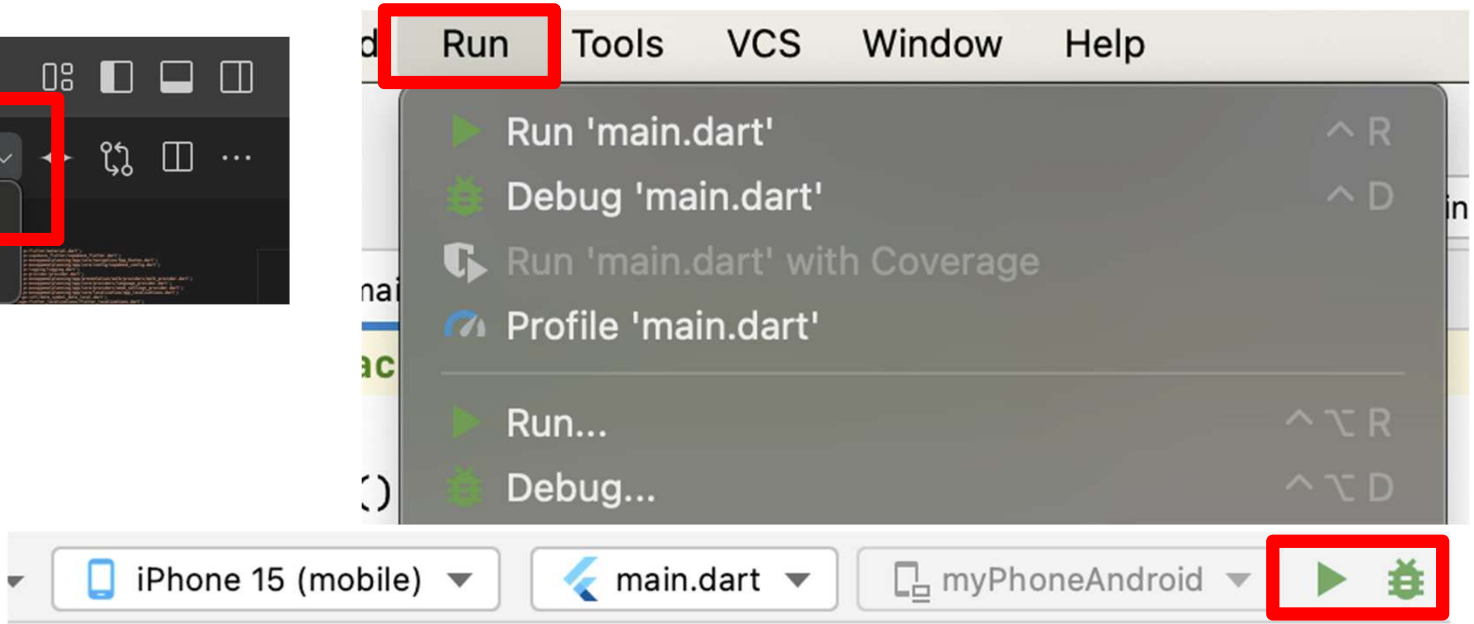
2. From the main.dart file:



3. From VS Code menu:



4. From Android Studio:



Chapter evaluation

Q1- Is it possible to develop iOS apps on a Windows machine ?

- ❑ Yes.
- ❑ No.

Q2- Which methods can be used to create a new Flutter project?

- ❑ Using the flutter create command.
- ❑ Using Visual Studio's project creation wizard.
- ❑ Using the Android Studio New Flutter Project wizard.
- ❑ Using the Xcode project creation wizard.

Chapter evaluation

Q3- In a Flutter project, where are the ".dart" files placed?

- ❑ lib folder.
- ❑ build folder.
- ❑ AndroidManifest folder.
- ❑ Pubspec folder.
- ❑ Gradle folder.

Abbreviations & glossaries

- **VSCode** (Visual Studio Code).
- **IDE** (Integrated Development Environment).
- **CLI** (Command Line Interface).
- **AVD** (Android Virtual Device).
- **SDK** (Software Development Kit) is a collection of building blocks (i.e. software tools, libraries, documentation, code samples, processes, and guides) used to develop efficiently software applications for specific platforms or frameworks.
- **Xcode** is a complete developer toolset for creating apps across all Apple platforms.

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